

FLYING FOODS

BUSINESS REQUIREMENTS DOCUMENT

Restaurant & Inventory Management System

Integrating POS, Pastel & Supplier Workflows with Kitchen, Inventory & Procurement

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1. Executive Summary

This Business Requirements Document defines a new Restaurant and Inventory Management System for Flying Foods. The system takes over every workflow that happens after the existing POS produces a receipt. The POS continues to handle cashiering, card payment, receipt printing, and sales reporting (sales per cashier, hourly breakdowns, payment types). The new system handles kitchen fulfillment, waiter service, inventory at three locations (Restaurant Stores, Restaurant Kitchen, in-flight Unit), procurement (Budget then PO then Supplier then GRN then Invoice), wastage and stock-take, inter-unit transfers, and budget reporting.

The system integrates outward to two external systems: the existing POS (inbound sale events) and Pastel (outbound stock movements for accounting). It also emails Purchase Orders to suppliers directly while keeping a complete in-system audit log of every PO sent.

1.1 Business Drivers

- Eliminate the paper-and-Excel inventory workflow and replace it with a single source of truth.
- Get the kitchen, the stores, the procurement function, and the accounting system speaking the same language in real time.
- Make supplier engagement traceable: every PO recorded, every GRN matched, every invoice reconciled.
- Surface real consumption (recipe-driven deductions plus wastage and adjustments) so Budget vs Actual reflects reality.
- Stop duplicating work between operations and accounting by syncing stock movements to Pastel automatically.

1.2 Automation Principles

The system replaces every physical handover, paper form, and parallel spreadsheet with an automated digital flow. The principles below govern the design.

- No manual receipt entry. The POS pushes (or our system pulls) the sale automatically and the kitchen ticket appears on the Kitchen Display the moment the sale lands.
- No paper GRN, PO or issue notes. Every procurement and stock document lives in the system.
- No parallel spreadsheets. The system is the single source of truth for inventory, procurement, and audit history; Pastel receives a synchronised feed of stock movements.
- No manual stock deduction for sales. The system deducts ingredients automatically using the active recipe version.
- No handwritten issue or transfer notes. Issues and transfers are created in the system at the point of action.
- Human action is reserved for decisions (approvals, exception handling, marking out-of-stock, marking ready/served), not for data re-entry.

1.3 Scope at a Glance

In Scope	Out of Scope
Kitchen Display, Chef workflow, extra-usage logging	Cashiering, payment processing, receipt printing (POS)
Waiter view, order delivery & served status	Sales-per-cashier and payment-type reporting (POS)
Inventory at Stores, Kitchen, and Unit	Customer-facing online ordering / delivery apps
Procurement: Budget → PO → Supplier email → GRN → 3-way match	Full general ledger / payroll / HR (Pastel)
Wastage, breakage, stock-take adjustments	Replacing or competing with the existing POS
Inter-unit transfers (Restaurant ↔ Unit)	Recipe-costing engine (Phase 2 candidate)
Menu & recipe management with versioning	
POS integration (sale events in)	
Pastel integration (stock movements out)	
Inventory & procurement reporting	

2. Stakeholders & Roles

The system serves the roles below. The Cashier role is no longer in scope. Cashiering remains in the existing POS. Each role has distinct permissions and maps to specific workflows captured in user stories in Section 7.

#	Role	Primary Responsibility
1	Chef	Sees orders appear on the Kitchen Display when the POS sale lands, prepares them, logs any in-the-moment extra ingredient usage, marks status In Preparation → Ready.
2	Waiter	Sees orders marked Ready on the Waiter view, delivers each order, and marks it Served. Can flag orders Returned.
3	Storekeeper	Owns the restaurant central stores. Receives deliveries, performs stock-takes, logs breakage and spoilage, monitors automatic deductions.
4	Receiving Officer	Drives procurement. Receives approved budgets, creates and sends POs to suppliers, enters GRNs at goods-in, matches invoices.
5	Unit Issuer (In-Flight Catering)	Submits stock requests against the restaurant stores on Tuesdays and Thursdays. Manages the in-flight catering Unit inventory.
6	Restaurant Issuer	Submits daily stock requests against the restaurant stores for the restaurant kitchen.
7	Manager / Finance	Approves Purchase Budgets, wastage above threshold, transfers above threshold. Reviews Budget vs Actual, GRN register, wastage reports.
8	System Administrator	Manages users, roles, product, supplier, and menu/recipe masters (with versioning); configures thresholds and reason codes; oversees integrations.

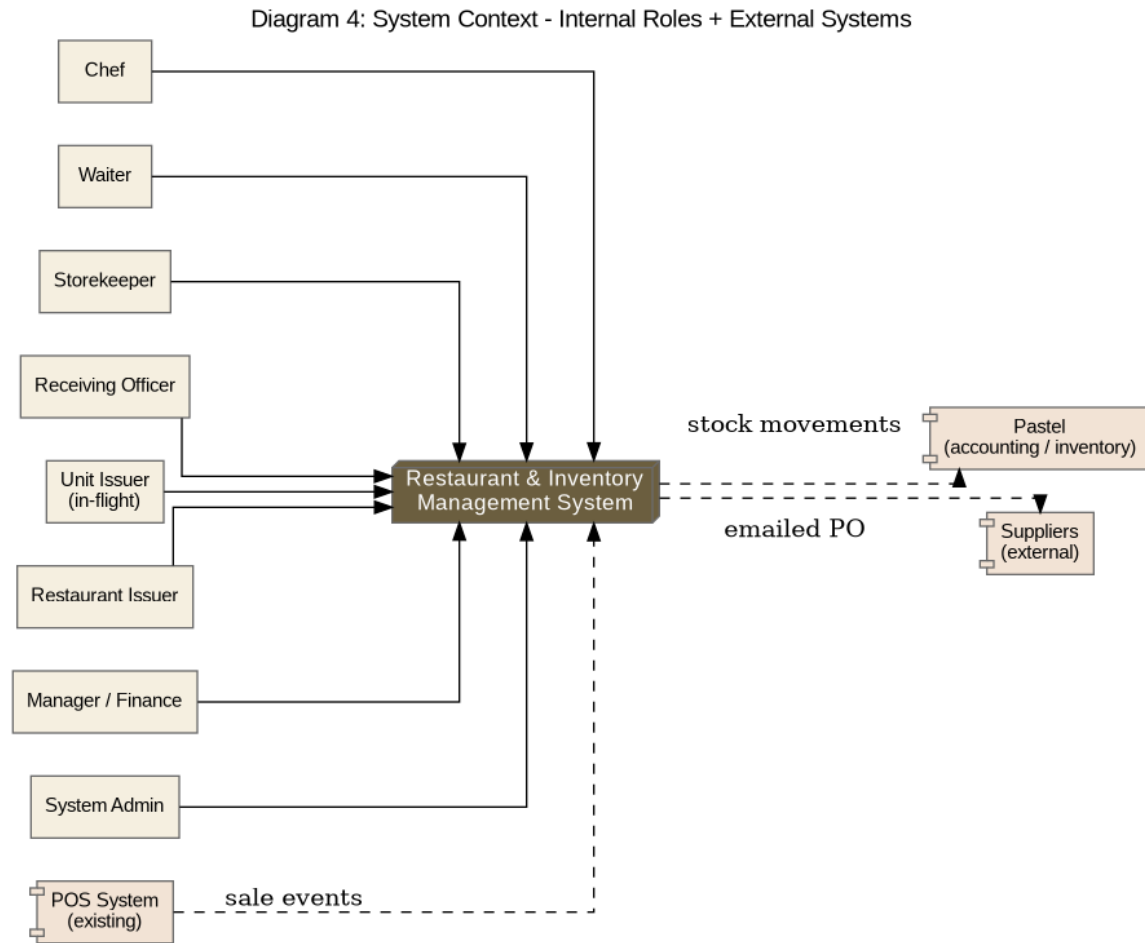


Figure 1. System context: internal roles plus the POS, Pastel, and supplier external systems

3. External System Integrations

The new system sits between the existing POS, the Pastel accounting/inventory system, and the supplier base. Two of these integrations have open design decisions that the stakeholders need to make before build. This section documents the options.

3.1 POS Integration (Inbound: Sale Events)

The existing POS handles cashiering, card payment, receipt printing, and its own sales reporting. The new system needs to know that a sale has happened so it can route the order to the kitchen and deduct ingredients. The question is how the sale gets from the POS to the new system.

Diagram 6: POS Integration Options - Push vs Pull

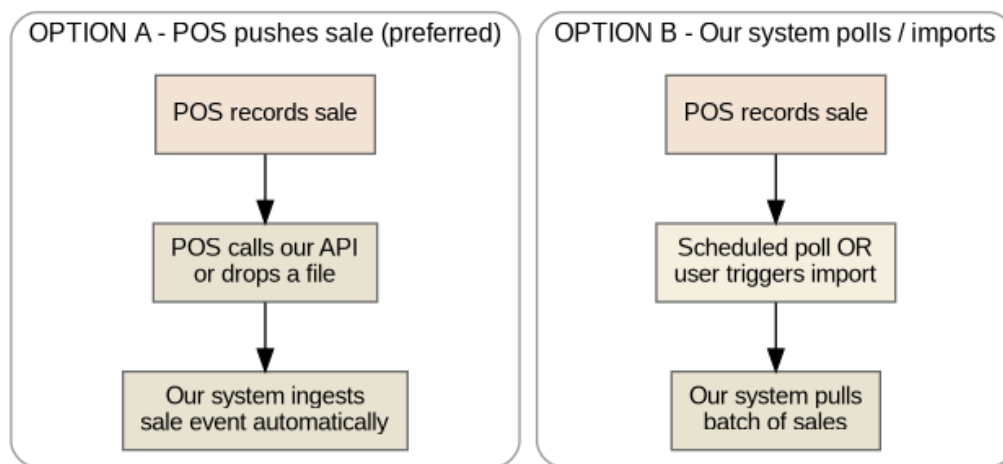


Figure 2. POS integration options: push (preferred) vs pull

Option	How it works	Pros	Cons
A. POS pushes (preferred)	POS calls our REST API or drops a file/event as soon as a sale completes.	Real-time kitchen ticket; no human action; matches the automation principles.	Requires POS vendor cooperation or POS-side scripting; depends on POS capability.
B. Our system pulls	Our system polls the POS database/API on a schedule, or a user imports a file.	Works even if POS cannot push; simpler to build on our side.	Latency between sale and kitchen ticket; user may need to trigger import; risk of duplicate or missed sales without careful watermarking.

Decision pending: stakeholders to confirm with the POS vendor which option is feasible. The new system shall be built to accept sale events through an internal API regardless, so that either option plugs into the same ingestion path.

3.2 Pastel Integration (Outbound: Stock Movements)

Pastel is the company's accounting/inventory system. To avoid double-entry and keep Pastel's inventory aligned with operational reality, the new system pushes stock movements to Pastel: goods received (GRN), stock issues, inter-unit transfers, wastage and stock-take adjustments, and recipe-driven sales deductions. The question is how live the sync should be.

Diagram 7: Pastel Sync Options - Real-time / Batch / Event

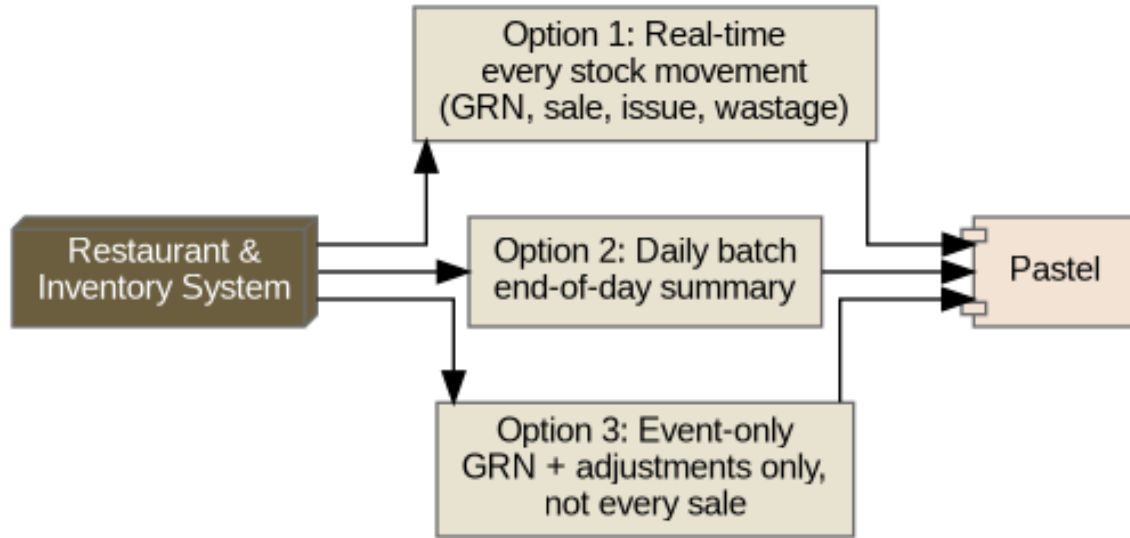


Figure 3. Pastel sync options: real-time / daily batch / event-only

Option	How it works	Pros	Cons
1. Real-time	Every stock movement (GRN, sale, issue, transfer, wastage) syncs to Pastel as it happens.	Pastel always matches operational reality; finance can run reports any time.	Highest integration complexity; many small Pastel transactions; needs robust retry on failure.
2. Daily batch	End-of-day job sends a summary of all stock movements for the day.	Simpler; one well-tested job per day; fewer Pastel calls.	Pastel lags operations by up to 24 hours; reconciliation is end-of-day only.
3. Event-only	Only GRNs and adjustments above a threshold are synced; recipe-driven sale deductions stay inside the new system.	Lightest integration; finance still sees the big movements.	Pastel's inventory will drift from operational on-hand; reports can disagree.

Decision pending: recommended option is Real-time with retry, with a Daily Batch fallback if Pastel's API cannot sustain the volume. To be confirmed in technical design with the Pastel administrator.

3.3 Supplier Integration (Outbound: Purchase Orders)

When a Purchase Order is created against an approved budget, the system emails the PO to the supplier directly and records the send (recipient, timestamp, PO PDF) in an in-system audit log. Manual fallback is

supported when a supplier has no email on file: the PO is still recorded and the Receiving Officer is prompted to contact the supplier outside the system.

- Supplier master stores name, contact name, email address, phone, and approval status.
- PO generation produces a PDF attached to the email and stored on the PO record.
- Email send results (delivered, bounced) are logged where the email gateway provides feedback.
- All sent POs are searchable by supplier, date, and PO number.

4. Business Process Overview

This section describes the core business processes the system supports. Each is illustrated with a flow diagram. Detailed functional requirements appear in Section 6 and user stories in Section 7.

4.1 Process A: Sale Capture → Kitchen Fulfillment → Service

A sale is recorded in the POS and the sale event arrives in the new system (push or pull, per Section 3.1). The system immediately routes an order ticket to the Kitchen Display and deducts the recipe ingredients from the restaurant stores Opening Balance, syncing the deduction to Pastel. The Chef updates the status to In Preparation, prepares the order, and logs any extra ingredient usage (e.g. a dropped egg) before marking the order Ready. The order then appears on the Waiter view; the Waiter delivers it and marks it Served. No physical handover and no manual data entry between any of these steps.

Diagram 1: Sale Capture (POS) -> Kitchen Fulfillment -> Service

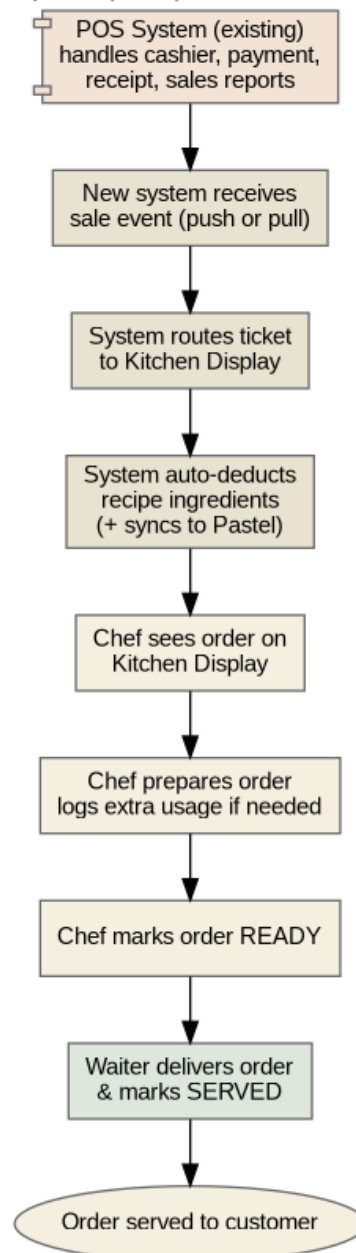


Figure 4. Process A: Sale capture, kitchen fulfillment & service

4.2 Process B: Procurement: Budget → PO → Supplier → GRN

A requester (Unit, Restaurant, or Stores) submits a Purchase Budget listing the items and quantities needed (e.g. "50 tomatoes"). The Manager reviews and approves on screen. Once approved, the Receiving Officer creates a Purchase Order against the approved budget, selecting one or more suppliers. The system emails the PO to each supplier and records the send. When the goods arrive, the Receiving Officer enters the GRN against the PO, capturing actual quantity received, unit cost, and any variance. The system increases the Opening Balance at Restaurant Stores, syncs the movement to

Pastel, and updates Budget vs Actual. The supplier invoice is later matched against the PO and GRN in a three-way match before payment is approved.

Diagram 2: Procurement - Budget -> PO -> Supplier -> GRN -> Pastel

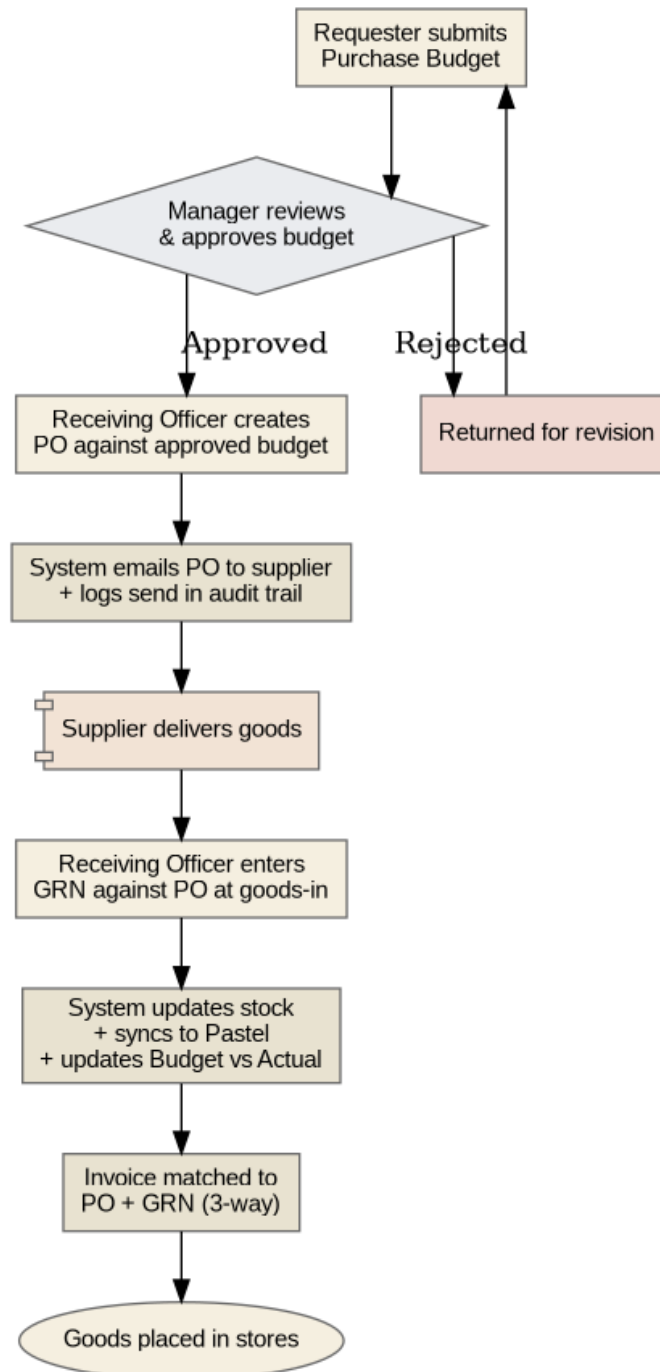


Figure 5. Process B: Full procurement flow from budget to invoice match

4.3 Process C: Inventory Issuing & Inter-Unit Transfers

The Restaurant Issuer creates issue requests against the restaurant stores every day. The Unit Issuer creates issue requests every Tuesday and Thursday. Each issue is recorded in the system, which deducts from central stores Opening Balance and adds to the destination location (Restaurant Kitchen or Unit). The Restaurant and Unit can also transfer stock between each other; transfers above a configurable threshold require Manager approval. Every movement is mirrored to Pastel per the integration option chosen.

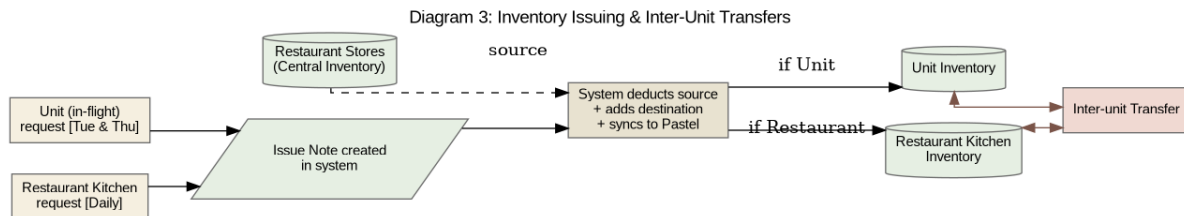


Figure 6. Process C: Inventory issuing & inter-unit transfers

4.4 Process D: Wastage, Breakage & Stock-Take

Recipe-driven auto-deduction handles the normal case, but real operations produce variances. Three event types feed the wastage ledger: in-the-moment extra usage during preparation (logged by Chef or Storekeeper against the active order); breakage or spoilage events (logged with a reason code); and periodic stock-take variances (physical count vs system on-hand). Adjustments above a configurable monetary threshold require Manager approval before posting to stock. All adjustments are mirrored to Pastel and feed into the Wastage Report and Budget vs Actual so real consumption is visible.

Diagram 5: Wastage, Adjustments & Stock-Take

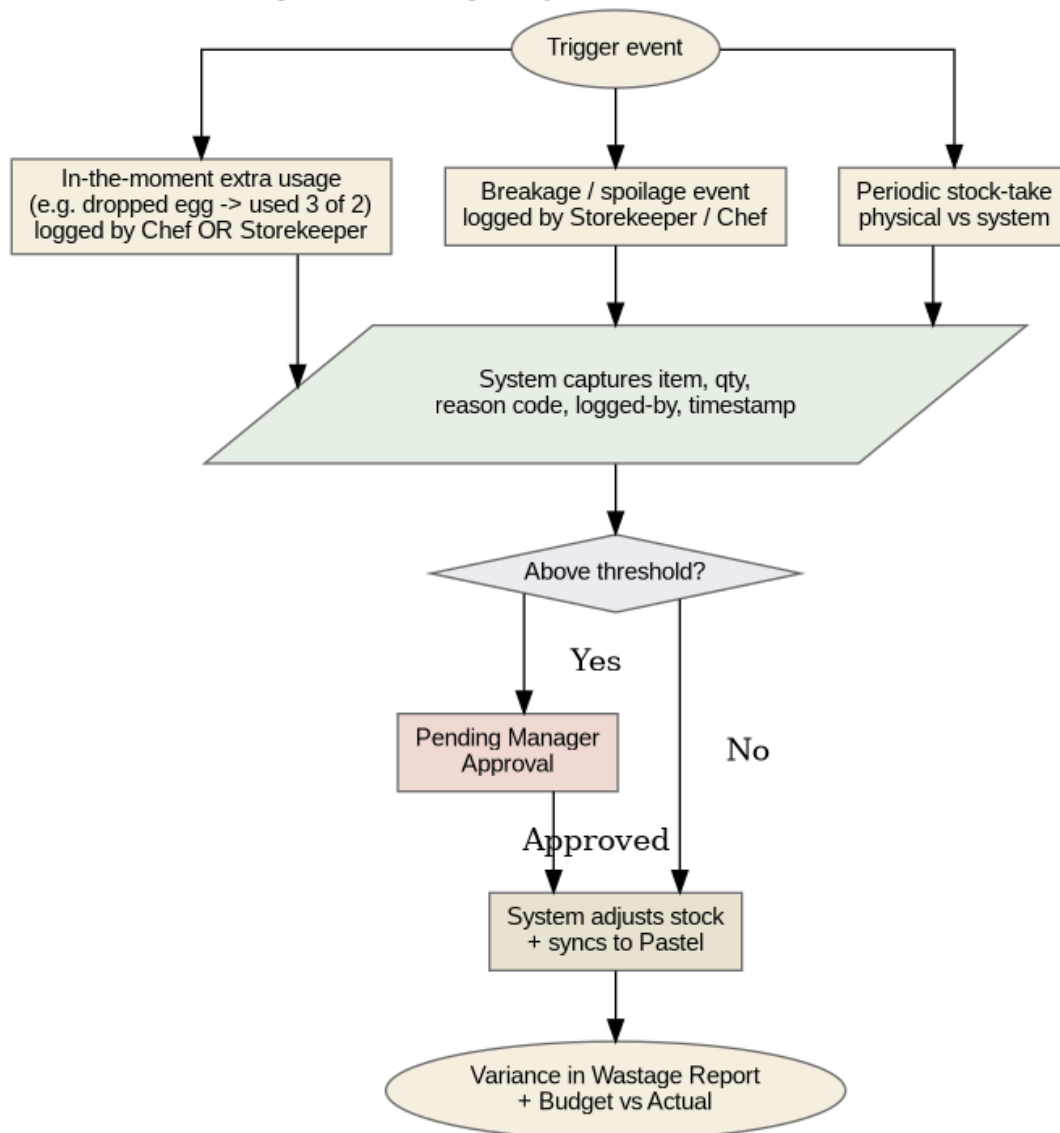


Figure 7. Process D: Wastage, adjustments & stock-take

4.5 Process E: Menu & Recipe Management

Menus and recipes change. The system uses versioned menu items and recipes with effective-from dates: editing a recipe creates a new version, never overwrites the previous one. Historical orders always reference the recipe version that was active when the order was placed, so Budget vs Actual and cost reports remain accurate over time. A Draft → Review → Published workflow lets the Administrator prepare changes (e.g. a seasonal menu) ahead of an effective date without affecting live service.

5. Assumptions, Constraints & Dependencies

5.1 Assumptions

- The existing POS will continue to be the system of record for cashiering, card payment, receipt printing, and sales-side reporting (sales per cashier, hourly, payment type).
- The POS can either push sale events or expose data we can pull (final integration approach pending vendor confirmation).
- Pastel exposes an API or import format suitable for posting inventory movements.
- Every menu item has a defined recipe so ingredient deduction from stock is automatic.
- The Restaurant operates a single central stores location; the Unit operates its own sub-inventory.
- Each role has access to a workstation, tablet, or shared device on the local network.

5.2 Constraints

- Cashiering and payment must remain in the existing POS; the new system must not duplicate them.
- Unit ordering is restricted to Tuesdays and Thursdays per business policy.
- The kitchen and stores areas must remain usable during short internet outages.

5.3 Dependencies

- Vendor cooperation from the POS supplier to confirm the integration approach (push API, file drop, or pull).
- Access to Pastel administrator for API credentials and chart-of-accounts mapping for stock movements.
- Current product master data, supplier list, and opening balances loaded into the new system at go-live.
- Defined recipes for every active menu item.

6. Functional Requirements

Functional requirements are grouped by module. Each has an ID (FR-xx-NN) and priority: M = Must have, S = Should have, C = Could have.

6.1 POS Integration Module (FR-PI)

ID	Requirement	Priority
FR-PI-01	The system shall accept sale events from the existing POS through an internal API supporting both push (POS-initiated) and pull (system-initiated) ingestion.	M
FR-PI-02	A sale event shall include sale ID, timestamp, cashier, line items (menu item, quantity, modifiers), and totals.	M
FR-PI-03	The system shall deduplicate sale events using the POS sale ID so the same sale is never ingested twice.	M
FR-PI-04	The system shall persist a raw copy of every ingested sale event for audit and replay.	M
FR-PI-05	If a sale event references an unknown menu item or recipe, the system shall ingest it but flag it for the Administrator and not deduct stock until resolved.	S
FR-PI-06	The system shall provide an admin screen to re-trigger or replay ingestion for a given sale ID, with audit log of who replayed it.	S

6.2 Pastel Integration Module (FR-PA)

ID	Requirement	Priority
FR-PA-01	The system shall push outbound stock movements to Pastel for GRNs, issues, transfers, wastage, stock-take adjustments, and sale-driven deductions.	M
FR-PA-02	The system shall support configurable sync mode: Real-time, Daily Batch, or Event-Only (per Section 3.2).	M
FR-PA-03	Failed syncs shall enter a retry queue with exponential backoff and shall be visible to the Administrator.	M
FR-PA-04	Every sync attempt (success or failure) shall be logged with timestamp, payload, and Pastel response.	M
FR-PA-05	The Administrator shall be able to manually re-send a failed sync after correction.	S
FR-PA-06	A daily Pastel Reconciliation Report shall list any item where our on-hand and Pastel's on-hand diverge.	S

6.3 Kitchen Display & Completion Module (FR-KD)

ID	Requirement	Priority
FR-KD-01	The system shall display all active orders to the Chef on the Kitchen Display in real time as soon as the POS sale event is ingested.	M
FR-KD-02	The system shall automatically deduct recipe ingredients from Restaurant Stores Opening Balance the moment the order reaches the Kitchen Display, and sync the deduction to Pastel.	M
FR-KD-03	The system shall allow the Chef to mark an order as In Preparation and then Ready. Marking Ready shall make the order visible on the Waiter view.	M
FR-KD-04	The system shall record a timestamp at each status change (Ingested, In Preparation, Ready, Served) for service-time reporting.	S
FR-KD-05	The system shall flag any order whose recipe ingredients are insufficient in stock and alert the Storekeeper.	S
FR-KD-06	The system shall allow the Chef to mark a menu item Out of Stock; out-of-stock items shall be flagged back to the POS if the POS supports a menu-availability feed (otherwise the Chef will communicate directly).	S
FR-KD-07	The system shall allow the Chef to log extra ingredient usage against an active order (e.g. dropped egg → used 3 of 2) before marking it Ready, with a reason code.	M

6.4 Waiter / Service Module (FR-SV)

ID	Requirement	Priority
FR-SV-01	The system shall provide a Waiter view that displays all orders in Ready status, with table or counter reference and time-since-Ready.	M
FR-SV-02	The system shall allow the Waiter to mark an order Served once delivered.	M
FR-SV-03	The system shall record the Waiter user, the table/counter, and the served timestamp on every served order for service-time reporting.	M
FR-SV-04	The system shall allow the Waiter to flag an order Returned with a reason, which alerts the Chef.	S

6.5 Procurement Module (FR-PR)

ID	Requirement	Priority
FR-PR-01	The system shall allow any authorised role to submit a Purchase Budget listing items and quantities.	M
FR-PR-02	The system shall route every Purchase Budget to the Manager for approval before purchase.	M

ID	Requirement	Priority
FR-PR-03	The system shall notify the Receiving Officer when a Purchase Budget is approved.	M
FR-PR-04	The Receiving Officer shall be able to create one or more Purchase Orders against an approved Purchase Budget, splitting items across suppliers as needed.	M
FR-PR-05	The system shall generate a PO PDF and email it to the supplier's email address on file, recording the send (recipient, timestamp, PDF) in an in-system log.	M
FR-PR-06	When a supplier has no email on file, the system shall still record the PO and prompt the Receiving Officer to contact the supplier outside the system; the PO status shall reflect this.	M
FR-PR-07	The Receiving Officer shall be able to enter a Goods Received Note (GRN) against a specific PO at goods-in, capturing item, quantity received, unit cost, condition, and variance reason if any.	M
FR-PR-08	On posting a GRN, the system shall increase Opening Balance at Restaurant Stores, sync to Pastel, and update Budget vs Actual.	M
FR-PR-09	The system shall support partial GRNs (multiple deliveries against one PO) with a running fulfilled-quantity per line.	S
FR-PR-10	The system shall support a three-way match between PO, GRN, and supplier Invoice; mismatches shall be flagged for review.	M
FR-PR-11	The system shall maintain a per-supplier history of POs, GRNs, invoices, and any disputes or returns.	S

6.6 Inventory: Issuing & Transfers Module (FR-IS)

ID	Requirement	Priority
FR-IS-01	The system shall allow the Restaurant Issuer to create daily Issue Notes from Restaurant Stores to Restaurant Kitchen.	M
FR-IS-02	The system shall allow the Unit Issuer to create Issue Notes from Restaurant Stores to the Unit on Tuesdays and Thursdays.	M
FR-IS-03	The system shall warn (but not block) the Unit Issuer when attempting to issue on a day other than Tuesday or Thursday.	S
FR-IS-04	The system shall deduct issued quantities from the source and add them to the destination location, and sync the movement to Pastel.	M
FR-IS-05	The system shall support inter-unit transfers between Restaurant Kitchen and Unit in both directions.	M
FR-IS-06	The system shall require Manager approval for inter-unit transfers above a configurable monetary threshold.	S

ID	Requirement	Priority
FR-IS-07	The system shall prevent posting an Issue Note that would drive any item's on-hand below zero, unless overridden by a Manager.	M

6.7 Wastage & Adjustments Module (FR-WA)

ID	Requirement	Priority
FR-WA-01	The system shall allow either the Chef or the Storekeeper to log extra ingredient usage against an active order, recording who logged it.	M
FR-WA-02	The system shall allow logging of breakage or spoilage events at any time, capturing item, quantity, reason code, and free-text note.	M
FR-WA-03	The system shall maintain a configurable list of reason codes for wastage.	M
FR-WA-04	The system shall support periodic stock-take entry, computing variance per item and posting it as an adjustment.	M
FR-WA-05	The system shall require Manager approval for wastage or adjustments above a configurable monetary threshold.	M
FR-WA-06	Approved wastage and adjustments shall be deducted from on-hand stock, written to the audit log, and synced to Pastel.	M
FR-WA-07	The system shall provide a Wastage Report broken down by reason, item, location, user, and period.	M
FR-WA-08	The system shall include wastage figures in Budget vs Actual.	S

6.8 Menu & Recipe Management Module (FR-MR)

ID	Requirement	Priority
FR-MR-01	The system shall maintain versioned menu items, each with effective-from and effective-to dates.	M
FR-MR-02	The system shall maintain versioned recipes; a recipe change creates a new version rather than overwriting the previous one.	M
FR-MR-03	For any historical order, the system shall apply the recipe version active on the date of that order.	M
FR-MR-04	The system shall support a Draft → Review → Published workflow for menu and recipe changes.	S
FR-MR-05	The system shall not allow deletion of a menu item or recipe with historical transactions; deactivation only.	M

ID	Requirement	Priority
FR-MR-06	The system shall allow scheduling a menu item to be available only on certain days or time-windows.	C

6.9 Master Data & Administration (FR-AD)

ID	Requirement	Priority
FR-AD-01	The Administrator shall be able to create, edit, and deactivate users and assign roles.	M
FR-AD-02	The system shall maintain a Product master (code, name, unit, category, reorder level).	M
FR-AD-03	The system shall maintain a Supplier master (name, contact, email, phone, approval status, payment terms).	M
FR-AD-04	The system shall maintain configurable approval thresholds for inter-unit transfers and wastage.	S
FR-AD-05	The system shall maintain configurable reason-code lists for wastage and returns.	S
FR-AD-06	The system shall maintain integration settings: POS endpoint, Pastel mode and credentials, supplier email gateway settings.	M

6.10 Reporting & Dashboards (FR-RP)

Reporting in this system focuses on inventory, procurement, and operational performance. Sales-side reporting (sales-per-cashier, hourly, payment type) remains in the POS and is intentionally not duplicated here.

ID	Requirement	Priority
FR-RP-01	The system shall provide a Stock-on-Hand report by location (Stores, Kitchen, Unit) for any selected date.	M
FR-RP-02	The system shall provide a Budget vs Actual report comparing approved budgets, GRN actuals, and wastage per period.	M
FR-RP-03	The system shall provide a PO Register by supplier, date, and status.	M
FR-RP-04	The system shall provide a GRN Register by date, supplier, item.	M
FR-RP-05	The system shall provide an Issues by Destination report (Unit vs Restaurant Kitchen).	S
FR-RP-06	The system shall provide a Wastage Report by reason, item, location, user, and period.	M
FR-RP-07	The system shall provide a Service Time report showing average and median Ingested → In Preparation → Ready → Served.	S

ID	Requirement	Priority
FR-RP-08	The system shall provide a Pastel Reconciliation Report flagging any divergence between our on-hand and Pastel.	S
FR-RP-09	All reports shall be exportable to PDF and Excel.	M

7. User Stories & Acceptance Criteria

Stories follow As a <role>, I want <capability> so that <business value>, with testable Given/When/Then acceptance criteria.

7.1 Chef Stories

US-CHF-01: Chef

As a **Chef**, I want **see every paid order on the Kitchen Display the moment the POS records it** so that **I can start preparing without waiting on anyone to tell me**.

Acceptance Criteria

- When the POS records a sale, the order appears on my Kitchen Display within 5 seconds of ingestion.
- Each order card shows order number, table or counter, items, modifiers, and time since ingestion.
- I never need to receive a paper ticket. The system is the channel.

US-CHF-02: Chef

As a **Chef**, I want **mark an order In Preparation and then Ready** so that **the Waiter and the counter see real-time order status**.

Acceptance Criteria

- Tapping Start changes status to In Preparation and records a timestamp.
- Tapping Ready moves the order to the Waiter view.
- A Ready order remains visible on my display until the Waiter marks it Served, so I can spot delays.

US-CHF-03: Chef

As a **Chef**, I want **log extra ingredient usage against an order when something goes wrong** so that **stock-on-hand stays accurate even when reality differs from the recipe**.

Acceptance Criteria

- When I add an extra-usage entry (item, quantity, reason), the system records who logged it.
- Entries below the threshold post immediately; above the threshold they go to Manager approval.
- I cannot mark the order Ready until a required approval is resolved or I withdraw the entry.

7.2 Waiter Stories

US-WTR-01: Waiter

As a **Waiter**, I want **see all orders that are Ready in one view** so that **I know what to deliver without walking back to ask**.

Acceptance Criteria

- My view lists every Ready order with table, items, and time-since-Ready, oldest first.
- A new Ready order appears within 5 seconds of the Chef marking it Ready.

US-WTR-02: Waiter

As a **Waiter**, I want **mark an order Served once delivered** so that **the team and Management can see real service status**.

Acceptance Criteria

- Tapping Served moves the order out of my view and records the timestamp against my user.
- The Service Time report can now calculate Ingested → InPrep → Ready → Served.

US-WTR-03: Waiter

As a **Waiter**, I want **flag an order Returned with a reason** so that **the kitchen can correct it and the issue is captured for review**.

Acceptance Criteria

- Tapping Return and selecting a reason sends the order back to In Preparation.
- My user and the reason are recorded against the order.

7.3 Storekeeper Stories**US-STK-01: Storekeeper**

As a **Storekeeper**, I want **have stock deducted automatically by the system when an order is placed** so that **I never manually adjust stock for sales and can focus on stock-take and exceptions**.

Acceptance Criteria

- When a sale reaches the Kitchen Display, the recipe's ingredients are deducted from Restaurant Stores with no action from me.
- A deduction that would go below zero triggers an alert to me.
- I can review every automatic deduction in an audit log and reverse one with a captured reason.

US-STK-02: Storekeeper

As a **Storekeeper**, I want **see real-time stock-on-hand by item** so that **I know what is low without counting manually**.

Acceptance Criteria

- Stock On Hand shows current quantities by item, sorted by category.
- Items at or below reorder level are highlighted.

US-STK-03: Storekeeper

As a **Storekeeper**, I want **log breakage, spoilage and other wastage events** so that **physical stock and system stock stay aligned**.

Acceptance Criteria

- I enter item, quantity, reason code, and optional note.
- Below threshold posts immediately; above threshold goes to Manager approval.
- Posted entries appear in the audit log and sync to Pastel.

US-STK-04: Storekeeper

As a **Storekeeper**, I want **perform a periodic stock-take and have variances posted as adjustments** so that **the system reflects reality and I can investigate any large variance**.

Acceptance Criteria

- A count sheet lists every active item with current on-hand.
- On submit, the system calculates variance per line and posts the total as a Stock-Take Adjustment.
- Above-threshold totals require Manager approval before posting.

7.4 Receiving Officer Stories**US-RCV-01: Receiving Officer**

As a **Receiving Officer**, I want **be notified when a Purchase Budget is approved** so that **I know exactly what to purchase and within what budget**.

Acceptance Criteria

- I receive an in-app notification with the budget number and total.
- I can open the budget and see all line items, requested quantities, and the approved monetary limit.

US-RCV-02: Receiving Officer

As a **Receiving Officer**, I want **create a Purchase Order against an approved budget and email it to the supplier** so that **procurement is traceable end-to-end and the supplier receives a clean document**.

Acceptance Criteria

- I pick a supplier from the master and select lines from the approved budget.
- On submit, the system generates a PO PDF and emails it to the supplier's email on file.
- The send (recipient, timestamp, PDF) is recorded in the PO's audit log.
- If the supplier has no email on file, the system records the PO and prompts me to contact them outside the system; the PO status reflects this.

US-RCV-03: Receiving Officer

As a **Receiving Officer**, I want **enter a GRN against a specific PO at goods-in** so that **goods received are traceable and stock is updated**.

Acceptance Criteria

- I select the PO, enter actual quantity, condition, and unit cost per line.
- On post, Opening Balance increases at Restaurant Stores, the movement syncs to Pastel, and Budget vs Actual updates.
- A variance vs the PO prompts me for a reason.

US-RCV-04: Receiving Officer

As a **Receiving Officer**, I want **match a supplier invoice against the PO and GRN** so that **we only pay for what we ordered and received**.

Acceptance Criteria

- I open the PO, see the matched GRN, and enter the invoice.
- The system flags any discrepancy between PO, GRN, and invoice.
- I can resolve, dispute, or refer the discrepancy to the Manager.

7.5 Unit Issuer (In-Flight Catering) Stories

US-UNT-01: Unit Issuer

As a **Unit Issuer**, I want **request stock from Restaurant Stores on Tuesdays and Thursdays** so that **the Unit has the ingredients it needs for the week**.

Acceptance Criteria

- On a Tue/Thu, my request is accepted normally.
- On other days, the system warns me and asks for a reason but lets me proceed.
- On post, quantities are deducted from Stores and added to Unit inventory, and the movement syncs to Pastel.

US-UNT-02: Unit Issuer

As a **Unit Issuer**, I want **transfer stock from the Unit to the Restaurant Kitchen** so that **we can help out when the Restaurant Kitchen runs short**.

Acceptance Criteria

- I create a transfer with Unit as source and Restaurant Kitchen as destination.
- Above the configured threshold, the transfer enters Pending Manager Approval before posting.

7.6 Restaurant Issuer Stories

US-RST-01: Restaurant Issuer

As a **Restaurant Issuer**, I want **request stock from Restaurant Stores every day** so that **the Restaurant Kitchen always has the ingredients needed for service**.

Acceptance Criteria

- I can create an Issue Request on any day.
- On post, quantities are deducted from Stores and added to Restaurant Kitchen inventory.

US-RST-02: Restaurant Issuer

As a **Restaurant Issuer**, I want **transfer stock from the Restaurant Kitchen to the Unit** so that **the Unit can be supported between its order days**.

Acceptance Criteria

- I create a transfer with Restaurant Kitchen as source and Unit as destination.
- Above threshold, it awaits Manager approval before posting.

7.7 Manager / Finance Stories

US-MGR-01: Manager

As a **Manager**, I want **approve or reject Purchase Budgets** so that **spending is controlled and traceable**.

Acceptance Criteria

- I receive a notification with full line item detail.
- I can Approve, Reject (with reason), or Request Revision.
- Approved budgets become available to the Receiving Officer.

US-MGR-02: Manager

As a **Manager**, I want **approve or reject wastage and stock-take adjustments above threshold** so that **large losses are not posted silently and I can investigate root cause**.

Acceptance Criteria

- I receive a notification with item, quantity, reason, value, and who logged it.
- I can Approve, Reject, or Request Investigation.
- I can see history of approved/rejected adjustments per period for trend analysis.

US-MGR-03: Manager

As a **Manager**, I want **see Budget vs Actual, including wastage** so that **I can monitor whether goods purchases stay within plan and how much is being lost to wastage**.

Acceptance Criteria

- The report shows approved budget, GRN actual, wastage value, and variance for a date range.
- Variances over a configurable percentage are highlighted.
- I can drill from any line into the underlying GRNs or wastage entries.

US-MGR-04: Manager

As a **Manager**, I want **see real-time stock at every location** so that **I can spot shortages or overstocking before they become a problem**.

Acceptance Criteria

- The dashboard shows current stock-on-hand at Stores, Kitchen, and Unit.
- Items below reorder level are flagged.

7.8 System Administrator Stories

US-ADM-01: System Administrator

As a **System Administrator**, I want **manage users and roles** so that **only authorised people can access the system**.

Acceptance Criteria

- I can create, edit, and deactivate users; deactivated users keep their history.

US-ADM-02: System Administrator

As a **System Administrator**, I want **maintain Product and Supplier masters** so that **every transaction references correct data**.

Acceptance Criteria

- I can add, edit, deactivate (not delete) Products and Suppliers.
- Supplier records include email used for PO sending.

US-ADM-03: System Administrator

As a **System Administrator**, I want **maintain versioned menu items and recipes** so that **menu changes do not corrupt historical reports**.

Acceptance Criteria

- Editing a recipe creates a new version with effective-from set to a date I choose.
- Historical orders continue to use the previous version.
- A Draft → Review → Publish workflow keeps in-progress changes out of live service.

US-ADM-04: System Administrator

As a **System Administrator**, I want **configure POS and Pastel integration settings and monitor them** so that **I can keep the integrations healthy and resolve failures quickly**.

Acceptance Criteria

- I can set POS ingestion mode (push or pull), endpoints, and credentials.
- I can set Pastel sync mode (real-time / batch / event-only).
- I can view the integration health dashboard and re-send failed syncs.

8. Non-Functional Requirements

Category	Requirement
Performance	Sale event to kitchen ticket on display: under 5 seconds end-to-end.
Availability	99% during operating hours. Local cached mode shall keep order taking, kitchen, and stores usable during short internet outages; sync resumes when connectivity returns.
Integration Resilience	POS ingestion and Pastel sync shall both retry on failure with exponential backoff and surface failures to the Administrator.
Security	Passwords hashed and salted. Role-based access on every screen and API endpoint. Integration credentials stored encrypted.
Audit	Every transaction logged with user, timestamp, and before/after values, including PO emails sent and Pastel syncs attempted.
Usability	Kitchen Display, Waiter view, and Stores screens usable on tablets and touch displays. Critical actions reachable in two taps.
Backup	Daily backups with documented restore procedure.
Localisation	Configurable currency, date format, and time zone.
Reporting	Reports export to PDF and Excel and paginate large outputs without freezing the UI.

9. High-Level Data Model

Key persisted entities. Field-level schemas defined in the Technical Specification.

Entity	Purpose	Key Attributes
User	System user account	user_id, name, email, role(s), status
Role	Permission grouping	role_id, name, permissions
Product	Inventory item	product_id, code, name, unit_of_measure, category, reorder_level, status, pastel_code
Menu Item	Versioned sellable item	menu_item_id, name, category, status
Recipe Version	Versioned BOM for menu item	recipe_version_id, menu_item_id, version_no, effective_from, effective_to, status
Recipe Line	Ingredient line in a recipe version	recipe_line_id, recipe_version_id, product_id, qty_per_serving
Supplier	External goods supplier	supplier_id, name, contact, email, phone, payment_terms, status
Location	Inventory holding location	location_id, name (Stores / Kitchen / Unit)
Stock Balance	Current on-hand per location	location_id, product_id, qty_on_hand, last_updated
Sale Event	Ingested POS sale	sale_event_id, pos_sale_id, ingested_at, payload, status
Order	Internal order created from sale event	order_id, sale_event_id, izwe_receipt_no, status, timestamps
Order Item	Line within an order	order_item_id, order_id, menu_item_id, recipe_version_id (snapshot), qty, modifiers, price
Order Status Event	Each status change	event_id, order_id, status, user_id, timestamp
Purchase Budget	Pre-approval purchase request	budget_id, items, total_estimated, status, requester, approver
Purchase Order	PO sent to supplier	po_id, budget_id, supplier_id, lines, status, pdf, sent_to, sent_at
GRN	Goods Received Note	grn_id, po_id, lines (product, qty, unit_cost), variance, posted_by
Invoice Match	3-way PO/GRN/Invoice match	match_id, po_id, grn_id, invoice_ref, status, discrepancies
Issue Note	Stock issue	issue_id, source_loc, dest_loc, lines, requester, status

Entity	Purpose	Key Attributes
Transfer	Inter-unit transfer	transfer_id, source_loc, dest_loc, lines, status, approver
Wastage Entry	Extra usage / breakage / spoilage	wastage_id, type, order_id (optional), product_id, qty, reason_code, logged_by, location, value, status
Stock-Take	Periodic physical count	stocktake_id, location_id, started_by, started_at, status, approver
Stock-Take Line	Count per item with variance	line_id, stocktake_id, product_id, system_qty, counted_qty, variance, value
Reason Code	Configurable reason list	code_id, category, label, active
Pastel Sync Log	Outbound sync attempts to Pastel	sync_id, entity, entity_id, attempted_at, status, payload, response
Audit Log	Append-only change history	log_id, entity, entity_id, user_id, action, before, after, timestamp

10. Reporting & Dashboards

Sales-side reporting (sales per cashier, hourly, payment type) remains in the POS. This system focuses on inventory, procurement, operations, and finance reconciliation.

Report / Dashboard	Primary Audience	Key Metrics
Stock On Hand	Storekeeper, Manager	Qty by item & location, items at/below reorder level
PO Register	Receiving Officer, Manager	POs by supplier, date, status, value
GRN Register	Receiving Officer, Manager	GRNs by date, supplier, item, cost, variance vs PO
Budget vs Actual	Manager / Finance	Approved budget, GRN actual, wastage value, variance & %
Wastage Report	Storekeeper, Chef, Manager	Wastage by reason, item, location, user, value and qty
Issues by Destination	Manager	Issues to Unit vs Restaurant Kitchen, weekly/monthly
Inter-Unit Transfers	Manager	Transfers Unit ↔ Restaurant, with approval status
Service Time	Manager	Avg/median Ingested → InPrep → Ready → Served, per shift
Pastel Reconciliation	Administrator, Finance	Any item where our on-hand and Pastel's on-hand diverge
Slow / Fast Movers	Manager	Items by movement velocity over period

11. Glossary

Term	Definition
POS	The existing point-of-sale system. Handles cashiering, card payment, receipt printing, and sales-side reporting. The new system consumes sale events from the POS.
Pastel	The company's accounting/inventory system. The new system pushes stock movements to Pastel so accounting and operations stay aligned.
Unit	The in-flight catering unit. Orders stock from Restaurant Stores on Tuesdays and Thursdays.
Restaurant Stores	Central inventory holding location supplying Restaurant Kitchen and Unit.
Restaurant Kitchen	The kitchen serving the dine-in restaurant. Orders stock from Restaurant Stores daily.
Opening Balance	The on-hand stock at a location; continuously updated by GRNs, issues, sales deductions, transfers, and wastage.
Purchase Budget	A pre-approved list of items and quantities the Receiving Officer is authorised to purchase.
PO (Purchase Order)	A formal order placed with a supplier against an approved budget. Emailed to the supplier and logged in the system.
GRN	Goods Received Note. A record of stock physically received from a supplier against a PO.
3-Way Match	Matching PO ↔ GRN ↔ Supplier Invoice before approving payment.
Issue Note	Record of stock issued internally from Stores to Kitchen or Unit.
Transfer	Inter-unit stock movement between Restaurant Kitchen and Unit.
Recipe	Bill of materials for a menu item. Versioned with effective dates.
Recipe Version	A specific point-in-time recipe definition. Historical orders always reference the version active on the order date.
Wastage	Stock consumed outside normal recipe-driven sale: dropped, spoiled, broken, end-of-day, theft/loss, or stock-take variance.
Stock-Take	Physical count of items at a location, compared to system on-hand; variance posted as an adjustment.
Reason Code	Configurable label on voids, wastage, returns, and transfers for categorised reporting.

12. Document Approval

This BRD requires sign-off from the stakeholders below before development begins.

Name	Role	Signature	Date
	Project Sponsor		
	Restaurant Manager		
	Finance Manager		
	Unit (In-Flight) Manager		
	IT / System Owner		
	POS Vendor Liaison		
	Pastel Administrator		